Lecture - 21

13/11/2025

Theorem Let W_1, W_2 be subspaces of a finite dimensional vector space. Then

$$\mathrm{Dim}_F(W_1+W_2) = \mathrm{Dim}_F(W_1) + \mathrm{Dim}_F(W_2) - \mathrm{Dim}_F(W_1 \cap W_2)$$

Proof

Observe that $\mathrm{Dim}_F(W_1 \cap W_2) \leq \mathrm{Dim}_F(W_1)$ and that $\mathrm{Dim}_F(W_1 \cap W_2) \leq \mathrm{Dim}_F(W_2)$

Let $l := \mathrm{Dim}_F(W_1 \cap W_2)$, $l+m := \mathrm{Dim}_F(W_1)$, $l+n := \mathrm{Dim}_F(W_2)$

Let $\{u_1, u_2, \cdots, u_l\}$ be a basis of $W_1 \cap W_2$. Enlarge this set to a basis of W_1, namely, $\{u_1, u_2, \cdots, u_l, u_{l+1}, \cdots, u_{l+m}\}$

Note: Note that $\{u_1, u_2, \cdots, u_l\}$ is a linearly independent subset of W_1 and therefore it can be enlarged to a basis of W_1.

Similarly, obtain a basis $\{u_1, u_2, \cdots, u_l, v_{l+1}, \cdots, v_{l+n}\}$ of W_2.

Now we shall prove that set

$\mathcal{B} = \{u_1, u_2, \cdots, u_l, u_{l+1}, \cdots, u_{l+m}, v_{l+1}, \cdots, v_{l+n}\}$ is a basis of $W_1 + W_2$

ⓐ $\underline{W_1 + W_2 = \mathrm{Span}(\mathcal{B})}$

Any element in W_1+W_2 is of the form $\omega_1+\omega_2$, where $\omega_1 \in W_1$ and $\omega_2 \in W_2$ and

$$\omega_1 \in \text{Span}\{u_1, u_2, \cdots, u_\ell, u_{\ell+1}, \cdots, u_{\ell+m}\} \in \text{Span}(\mathcal{B})$$

and

$$\omega_2 \in \text{Span}\{u_1, u_2, \cdots, u_\ell, v_{\ell+1}, \cdots, v_{\ell+n}\} \in \text{Span}(\mathcal{B}).$$

Thus $\omega_1+\omega_2 \in \text{Span}(\mathcal{B})$ and this proves

$$W_1+W_2 \subseteq \text{Span}(\mathcal{B}).$$

Similarly one proves $\text{Span}(\mathcal{B}) \subseteq W_1+W_2$ (H.W.)

(b) $\mathcal{B}$ is a linearly independent set

Suppose that

$$\alpha_1 u_1+\alpha_2 u_2+\cdots+\alpha_\ell u_\ell+\alpha_{\ell+1}u_{\ell+1}+\cdots+\alpha_{\ell+m}u_{\ell+m}+\beta_{\ell+1}v_{\ell+1}+\cdots+\beta_{\ell+n}v_{\ell+n}=0$$

Then,

$$\alpha_1 u_1+\alpha_2 u_2+\cdots+\alpha_\ell u_\ell+\alpha_{\ell+1}u_{\ell+1}+\cdots+\alpha_{\ell+m}u_{\ell+m}=-\beta_{\ell+1}v_{\ell+1}+\cdots+-\beta_{\ell+n}v_{\ell+n}$$

Note that the expression in blue belongs to W_1, whereas, the expression in green belongs to W_2.

Therefore

$$-\beta_{\ell+1}v_{\ell+1}+\cdots+-\beta_{\ell+n}v_{\ell+n} \in W_1 \cap W_2 = \text{Span}\{u_1, \cdots, u_\ell\}.$$

Since the set $\{u_1, \cdots, u_\ell, v_{\ell+1}, \cdots, v_{\ell+n}\}$ is linearly independent,

$$\beta_{\ell+1}=0, \quad \beta_{\ell+2}=0, \quad \cdots, \quad \beta_{\ell+n}=0.$$

Now this implies

$$\alpha_1 u_1+\alpha_2 u_2+\cdots+\alpha_\ell u_\ell+\alpha_{\ell+1}u_{\ell+1}+\cdots+\alpha_{\ell+m}u_{\ell+m}=0$$

Now since the set $\{u_1, u_2, \dots, u_\ell, u_{\ell+1}, \dots, u_{\ell+m}\}$ is linearly independent, $\alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_{\ell+m} = 0$.

Thus $\mathcal{B}$ is a linearly independent set. Hence $\mathcal{B}$ is a basis of $W_1 + W_2$.

Finally, observe that

$$\begin{aligned}\mathrm{Dim}_F(W_1+W_2) &= \ell+m+n \\ &= \ell+m+\ell+n-\ell \\ &= \mathrm{Dim}_F(W_1) + \mathrm{Dim}_F(W_2) - \mathrm{Dim}_F(W_1 \cap W_2) \ \square\end{aligned}$$

<u>The Rank-Nullity theorem</u>, <u>Sylvester's law</u>, <u>First isomorphism theorem</u>

Let $T: V \longrightarrow W$ be a linear transformation of finite dimensional F-Vector Spaces. Then,

$$\mathrm{Dim}_F(\ker T) + \mathrm{Dim}_F(\mathrm{im}\, T) = \mathrm{Dim}_F(V).$$

<u>Note:</u> $\ker(T)$ is often called null space and $\mathrm{im}(T)$ is called Range of T, denoted by $\mathrm{Range}(T)$.

<u>Proof</u> Let $\{u_1, \dots, u_m\}$ be a basis of $\ker T$. Enlarge

this basis of $\ker T$ to a basis

$$\{u_1, u_2, \ldots, u_m, v_1, \ldots, v_n\}$$

of V. Thus

$$\mathrm{Dim}_F(\ker T) = m, \quad \mathrm{Dim}_F(V) = m+n.$$

Therefore, we only need to show $\mathrm{Dim}_F(\mathrm{im}\, T) = n$.

We will in fact show that $\{T(v_1), T(v_2), \ldots, T(v_n)\}$ is a basis of V.

$\{T(v_1), T(v_2), \ldots, T(v_n)\}$ spans $\mathrm{im}(T)$.

Let $w \in \mathrm{im}\, T$. Then, $w = T(v)$ and we may write

$$v = \alpha_1 u_1 + \cdots + \alpha_m u_m + \beta_1 v_1 + \cdots + \beta_n v_n.$$

$$\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n \in F$$

Observe that

$$\begin{aligned} w = T(v) &= T(\alpha_1 u_1 + \cdots + \alpha_m u_m + \beta_1 v_1 + \cdots + \beta_n v_n) \\ &= \alpha_1 T(u_1) + \cdots + \alpha_n T(u_m) + \beta_1 T(v_1) + \cdots + \beta_n T(v_n) \\ &= 0 + \beta_1 T(v_1) + \cdots + \beta_n T(v_n). \end{aligned}$$

Thus $\mathrm{im}\, T = \mathrm{Span}\{T(v_1), \ldots, T(v_n)\}$.

$\{T(v_1), T(v_2), \ldots, T(v_n)\}$ is a linearly independent set

If $\gamma_1 T(v_1) + \gamma_2 T(v_2) + \cdots + \gamma_n T(v_n) = 0$ for some $\gamma_1, \ldots, \gamma_n \in F$

then $\quad T(\gamma_1 v_1 + \gamma_2 v_2 + \cdots + \gamma_n v_n) = 0$

and therefore, $\quad \gamma_1 v_1 + \gamma_2 v_2 + \cdots + \gamma_n v_n \in \ker(T)$.

This implies

$$\gamma_1 v_1 + \gamma_2 v_2 + \cdots + \gamma_n v_n = \alpha_1 u_1 + \cdots + \alpha_m u_m$$

Then,

$$\gamma_1 v_1 + \gamma_2 v_2 + \cdots + \gamma_n v_n - \alpha_1 u_1 + \cdots + -\alpha_m u_m = 0$$

and since $\{u_1, \ldots, u_m, v_1, \ldots, v_n\}$ is a basis of V,

we must have $\boxed{\gamma_1 = 0,\ \gamma_2 = 0,\ \ldots,\ \gamma_n = 0}$, $\alpha_1 = 0, \ldots, \alpha_m = 0$.

$\boxed{\text{This shows } \{T(v_1), T(v_2), \ldots, T(v_n)\} \text{ is a basis of } V.}$ $\square$

Applications

① Let $T: \mathbb{R}^n \longrightarrow \mathbb{R}$ be a nonzero linear Transformation. What is the dimension of $\ker T$? (Recall HW Problem)

Since T is nonzero transformation, $\operatorname{im}(T) = \mathbb{R}$ (why)

That is, T is onto!

and we have

$$\operatorname{Dim}_{\mathbb{R}}(\mathbb{R}^n) = \operatorname{Dim}_{\mathbb{R}}(\ker T) + \operatorname{Dim}_{\mathbb{R}}(\operatorname{im}(T))$$

$$\Longrightarrow \quad n = \operatorname{Dim}_{\mathbb{R}}(\ker T) + 1$$

$$\Longrightarrow \quad \boxed{\operatorname{Dim}_{\mathbb{R}}(\ker T) = n - 1}$$

(2) Let $T: V \longrightarrow W$ be a linear transformation of (Second proof) finite dimensional vector spaces with $Dim_F(V) = n$, $Dim_F(W) = m$ and that $n > m$. Then T is <u>not</u> one to one (injective).

<u>Proof</u>

We know that $Dim_F(V) = Dim_F(\ker T) + Dim_F(\operatorname{im} T)$

Since $Dim_F(\operatorname{im} T) \leq m$, $\quad n \leq Dim_F(\ker T) + m$

$$\Rightarrow \quad Dim_F(\ker T) \geq n - m \geq 1.$$

Thus T is not one-one.

(3) Let $T: V \longrightarrow W$ be a linear transformation of (Second proof) finite dimensional vector spaces with $Dim_F(V) = n$, $Dim_F(W) = m$ and that $n < m$. Then T is <u>not</u> onto (Surjective)

H.W.